A Verbatim-Grounded, Field-Neutral Tag Layer for Cross-Disciplinary Reading of Academic Literature: A Proof of Format

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In three sentences. Researchers cannot find the work in other fields that studies the same thing as their own, because each field calls it by a different name. This paper describes a tag layer that gives every paper the same small set of tags and forces each tag to quote the exact sentence it came from — then reports honestly what that does and does not buy: free-text names for a phenomenon almost never collide across fields, while a controlled vocabulary of statistical terms does create real cross-field joins. The same instrument, read in reverse, can point at regions of concept space that the literature occupies but has never named.

A note on two numbers. The audited proof-of-format below is the frozen 126-paper corpus; the working corpus has since grown to 187 papers across 40 disciplines, including a digital-humanities batch added as a deliberate stress test (§8.1). Audited figures are not restated for the larger set, because they would have to be re-audited to be true.

Abstract. Academic knowledge is written almost entirely in natural language, which makes it readable by humans but hard to query *across* disciplines that use different words for related things. We describe a tag layer that assigns every paper the same small set of **field-neutral facets** — claim type, phenomenon (a normalised join key), claim relations, replication status, and limitations — under one non-negotiable design rule: **every tag must be licensed by an exact, verbatim sentence from the paper's own text** — a requirement this audit measures against the legacy data rather than assuming it was met. We report a proof-of-*format* on a curated corpus of 126 papers spanning 39 disciplines (1866–2023), of which 116 were fully faceted. The format is feasible and verifiable: 127 claim-relations and 87 limitations, each substring-checked against the source. We are equally explicit about what it did *not* do: a novelty scan surfaced no cross-disciplinary link that the literature had not already drawn, and the phenomenon facets, written as specific verbatim-anchored strings, rarely collide across fields — indicating that a **canonical join vocabulary**, not free text, is the missing ingredient for surfacing new links. We argue the value of the layer is as a **compass for "far reading"** that points to where slow close reading may be worthwhile, not as an answer engine. We then validate the fix the null result points to: a small controlled vocabulary of statistical terms supplies shared cross-field join keys, with the broadest audited term spanning **31 disciplines** (§5). A complete re-audit of 519 legacy paper–term assignments rejected 99 (19.1%) and retained 420 semantically accepted assignments; because source availability and exact excerpt grounding are incomplete, the primary counts and a stricter exact-local-body sensitivity analysis are reported separately. As a first structural test, 16 papers across 8 disciplines already group under one canonical

phenomenon (priming) despite each field naming it differently. A grounded audit of that cluster then asks the harder question — does each field's priming actually carry the definition's three features? — and finds **ASSOCIATION** and **MODULATION** verbatim-grounded in 16/16 papers and **SECONDARINESS** in 14/16, the two exceptions being genuine boundary findings (§5.1, Figs. 5–6). Two further results follow. Read in reverse, the null result of §4 becomes a **DETECTOR**: a region of concept space that is populated but carries no shared label marks a term the literature never coined. Applied to the memory taxonomies, the cell that would be called **SHORT-TERM IMPLICIT MEMORY** is crowded with data and almost nameless: roughly **4,350** works occupy it under five paradigm-specific names while *short-term implicit memory* and its variants appear in about **52** — a gap that is countable rather than impressionistic (§5.3). Finally, the reliability of the tagging is now **MEASURED** rather than asserted: an independent, blinded second coder re-coded 120 assignments, giving **κ = 0.767** (95% CI 0.652–0.882), with the disagreement asymmetric in the direction of the original coding being stricter (§7).

This is the complete paper, with every paragraph printed in full. The short version — one written sentence per paragraph, the same figures — is deposited beside this file, and the live page at shir-openu.github.io/meta-tagging-showcase/preprint.html shows those sentences with each paragraph one click behind its own.

Researchers cannot find the work in other fields that studies the same thing as their own, because each field calls it by a different name.

1. Motivation

Most scholarly literature is natural-language prose. Cross-disciplinary work is therefore bottlenecked not by a lack of relevant results elsewhere, but by the difficulty of *finding* them: the same mechanism, method, or phenomenon appears under different names in different fields. Existing structured resources — ontologies, citation graphs, keyword indexes — either require heavy per-field curation or capture topic words that are, by construction, discipline-specific and so useless as join keys.

Describe every paper with the same small set of tags, and require each tag to quote the sentence it came from.

2. The idea

We tag each paper with a fixed, deliberately small set of **facets chosen to be field-neutral** (they describe the *shape* of a contribution, not its topic): `claim-type` (existence / mechanism / method / prediction / critique / null-result), `phenomenon` (a normalised name intended as a cross-field join key), `claim-relations` (builds-on / extends / contradicts / replicates, each toward a named target), `replication-status`, and `limitations`. The defining constraint is **verbatim grounding**: a tag is an interpretation of natural language, so the design requires every non-trivial tag to carry a licensing excerpt and a resolvable provenance state. The present audit reports where the legacy data meets that requirement and where it does not. This is what separates the intended layer from an ungrounded ontology (whose assertions cannot be checked against a source) and from a dense embedding (which is not human-auditable). The layer serves humans (HTML views with the licensing sentence attached) and machines (a single JSON record per paper) at once.

One graph: papers and tags as nodes, each tag-to-paper edge carrying the exact words that justify it.

2.1 The data structure: one property graph over a grounded tag layer

The layer can be represented as a **property graph**, built in three layers (Fig. 1). At the base is the **corpus** — 126 papers as raw, natural-language text, one record each. Over it sits the **grounded tag layer**: every paper carries the same field-neutral facets, and each non-trivial tag stores its licensing excerpt and provenance status. The implemented graph export reads these records as a graph — **nodes** are papers, tags, disciplines, and canonical phenomena; **edges** are `HAS_TAG` (carrying its evidence), `IN` (paper→discipline), `ALIAS_OF` (a field's local name→a canonical phenomenon), `CONGRUENT_TO` (a scored 0–1 correspondence between two phenomena), and projected cross-field links. The current export implements `IN`, `HAS_CLAIM`, `HAS_PHENOMENON`, and audited `HAS_STATISTIC` edges; `ALIAS_OF`, `CONGRUENT_TO`, and projected links remain a proposed extension, not a measured output.

Two properties support scaling. First, verification becomes a **set query** rather than manual reading: “does each priming phenomenon carry association, secondariness and modulation?” is the intersection of four tag-sets, and its *complement* lists the exceptions automatically (§5.1). Second, the implementation maps each facet value to a sorted postings list. Dictionary lookup is expected O(1), while an ALL query intersects lists in $\Theta(\Sigma|P_i|)$ time plus output work; a common tag, unfiltered listing, or complement can therefore still be $\Theta(N)$. Selective queries scale by avoiding unrelated records, not by becoming independent of corpus size. Standard graph algorithms may run over the exported adjacency list with their ordinary graph-dependent costs (for example, $O(|V|+|E|)$ for a full traversal). The same query syntax can be used on thousands of papers, but runtime depends on postings lengths and graph size (§5.1, Fig. 6).

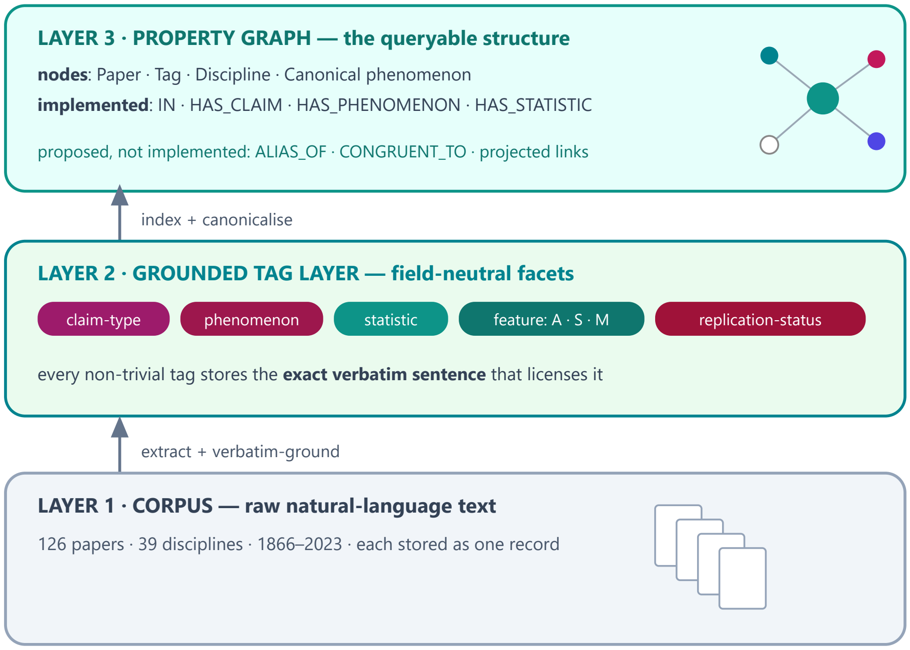


Fig. 1. The data structure: one **property graph** in three layers. A corpus of raw text (Layer 1) carries a grounded tag layer (Layer 2), which is read as a graph of nodes and edges (Layer 3) over which verification becomes a query. Full specification in the project's `GRAPH_SCHEMA`.

Concretely (Fig. 2), each paper is one **record** — a JSON document holding its facets and, for each audited statistical tag, the evidence excerpt, decision, reason, role, and provenance hashes. From these records the reproducible build derives two classic structures: an **inverted index** (a dictionary from a facet value to a sorted postings list of numeric document identifiers) and a typed **adjacency list** containing only relations present in the records. Multi-tag search is a sorted-list intersection or union; graph analyses can consume the adjacency export.

① one paper = one record (document store — a dict)

```
{
  "id": "newman2003-complex-networks",
  "discipline": "network-science", "year": 2003, "license": "arxiv-open",
  "title": "The Structure and Function of Complex Networks",
  "phenomenon": "complex networks (shared structure of real networks)",
  "content_tags": {
    "statistics": [{ "term":"correlation", "symbol":"r",
        "evidence":"For the data of Table III we find r = 0.621." }],
    "key_terms":  ["networks","degree","clustering","vertices","edges"] } }
```

② inverted index (facet → sorted postings) — expected O(1) key lookup

```
"statistic:correlation"      → [ 7, 12, 18, ... ] (39 papers · 22 disciplines)
"claim:method"               → [ 1, 3, 6, ... ]                  (55 papers)
"discipline:network-science" → [ 83, 84 ]                        (2 papers)
ALL = intersect; ANY = union; complement = universe difference
merge cost = Θ(Σ|P_i|) + output; worst case Θ(N)
```

③ implemented adjacency export (node → typed neighbours)

```
paper:newman2003 → [ IN:discipline:network-science,
                     HAS_STATISTIC:statistic:correlation, ... ]
not exported: ALIAS_OF · CONGRUENT_TO · projected links
```

Fig. 2. Implemented data structures. Each paper is one canonical JSON record. The build derives sorted postings and lazy metadata shards for search, plus a typed adjacency export. Dictionary lookup is expected O(1), but merging postings depends on their lengths and can be $\Theta(N)$; no corpus-size-independent complexity is claimed.

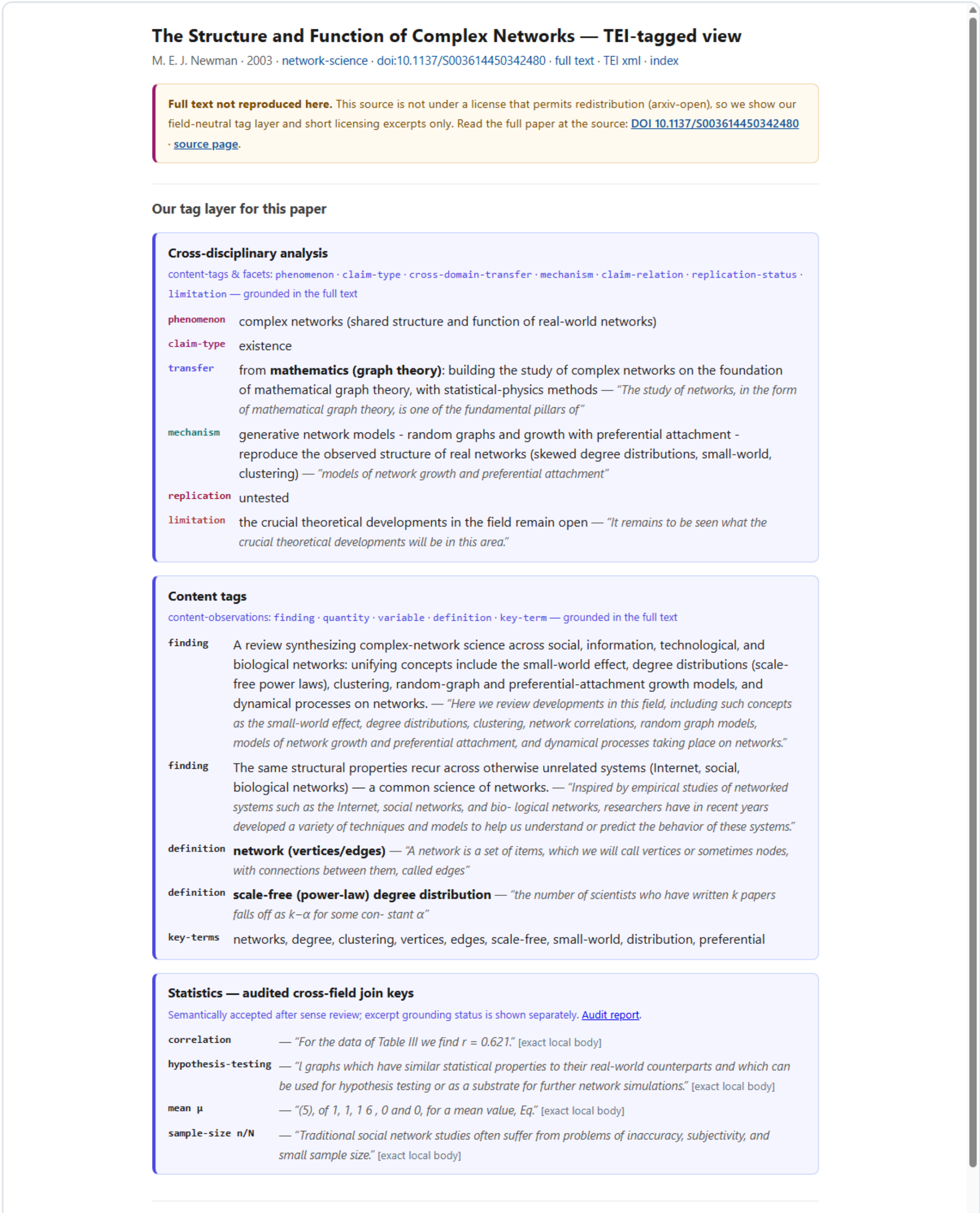
A curated corpus spanning dozens of disciplines, tagged to one schema, with every tag checked against the paper's own text.

3. Corpus and method

The corpus is a curated, deliberately broad sample — an *existence proof of the format and procedure, not a claim of completeness or representativeness*. Each paper is stored as a single canonical record; the implemented build audits and validates statistical annotations, recalculates the reported quantities, derives search shards and postings, exports a typed adjacency list, and regenerates the audit and paper views. The original faceting process intended character-level grounding, but the present re-audit shows that this invariant was not consistently preserved. Grounding status and hashes are now explicit fields, and unverified excerpts are reported rather than silently treated as exact.

Data quality. Of 126 papers, **10 (7.9%)** could not be faceted — all because of a *garbled source text layer* (image-only scans, dropped characters, interleaved sources), not because their content resisted the method. On clean text the method faceted 116/116. This failure rate therefore measures digitisation/sourcing quality, not the tagging method; we record such papers under an explicit `untaggable` tag rather than dropping them silently.

How a record is read back. Every open-access paper is rendered as a **colour-tagged reading view** (Fig. 3): the same facets shown inline over the paper's own text, each tag colour-coded and labelled with its type, so a human can audit every tag against the sentence that licenses it. Paywalled papers are linked by DOI, not reproduced.


The Structure and Function of Complex Networks — TEI-tagged view

M. E. J. Newman · 2003 · network-science · doi:10.1137/S003614450342480 · full text · TEI xml · index

Full text not reproduced here. This source is not under a license that permits redistribution (arxiv-open), so we show our field-neutral tag layer and short licensing excerpts only. Read the full paper at the source: DOI 10.1137/S003614450342480 · source page.

Our tag layer for this paper

Cross-disciplinary analysis

content-tags & facets: phenomenon · claim-type · cross-domain-transfer · mechanism · claim-relation · replication-status · limitation — grounded in the full text

phenomenon	complex networks (shared structure and function of real-world networks)
claim-type	existence
transfer	from **mathematics (graph theory)**: building the study of complex networks on the foundation of mathematical graph theory, with statistical-physics methods — *"The study of networks, in the form of mathematical graph theory, is one of the fundamental pillars of"*
mechanism	generative network models - random graphs and growth with preferential attachment - reproduce the observed structure of real networks (skewed degree distributions, small-world, clustering) — *"models of network growth and preferential attachment"*
replication	untested
limitation	the crucial theoretical developments in the field remain open — *"It remains to be seen what the crucial theoretical developments will be in this area."*

Content tags

content-observations: finding · quantity · variable · definition · key-term — grounded in the full text

finding	A review synthesizing complex-network science across social, information, technological, and biological networks: unifying concepts include the small-world effect, degree distributions (scale-free power laws), clustering, random-graph and preferential-attachment growth models, and dynamical processes on networks. — *"Here we review developments in this field, including such concepts as the small-world effect, degree distributions, clustering, network correlations, random graph models, models of network growth and preferential attachment, and dynamical processes taking place on networks."*
finding	The same structural properties recur across otherwise unrelated systems (Internet, social, biological networks) — a common science of networks. — *"Inspired by empirical studies of networked systems such as the Internet, social networks, and bio- logical networks, researchers have in recent years developed a variety of techniques and models to help us understand or predict the behavior of these systems."*
definition	**network (vertices/edges)** — *"A network is a set of items, which we will call vertices or sometimes nodes, with connections between them, called edges"*
definition	**scale-free (power-law) degree distribution** — *"the number of scientists who have written k papers falls off as k−α for some con- stant α"*
key-terms	networks, degree, clustering, vertices, edges, scale-free, small-world, distribution, preferential

Statistics — audited cross-field join keys

Semantically accepted after sense review; excerpt grounding status is shown separately. Audit report.

correlation	— *"For the data of Table III we find r = 0.621."* [exact local body]
hypothesis-testing	— *"l graphs which have similar statistical properties to their real-world counterparts and which can be used for hypothesis testing or as a substrate for further network simulations."* [exact local body]
mean μ	— *"(5), of 1, 1, 1 6 , 0 and 0, for a mean value, Eq."* [exact local body]
sample-size n/N	— *"Traditional social network studies often suffer from problems of inaccuracy, subjectivity, and small sample size."* [exact local body]

Fig. 3. The colour-tagged reading view (Newman 2003, an open-access paper): a colour legend plus the paper's own text with inline, type-labelled tags (persName, date, title, key-term, statistic…), each auditable against its sentence. **Click the figure** for the live page: shir-openu.github.io/meta-tagging-showcase/papers/newman2003….

The honest null result: free-text phenomenon names almost never collide across fields, so as join keys they fail.

4. Results

116 papers across 38 disciplines were fully faceted.

claim-type	method 55 · existence 42 · mechanism 9 · critique 8 · prediction 2 · null-result 0
replication-status	untested 113 · replicated 3 · (failed/contested 0 — see §7)
grounded relations	127 claim-relations, 87 limitations — all substring-verified

The central honest finding. As free-written strings, the 113 distinct `phenomenon` values almost never coincide across fields: only **1** pair of disciplines shares a phenomenon string verbatim. The bridging one might hope for (e.g. a "priming" or "power-law" motif recurring across many fields) is real at the level of shared *concepts*, but the strict, verbatim-anchored phenomenon facet is worded too specifically to collide. The implication is concrete: the join key needs a **canonical vocabulary** (a controlled, normalised phenomenon namespace) layered over the grounded free text — not more papers. A separate novelty scan, checking every cross-field candidate against the existing literature, found **no** connection that had not already been drawn; we report this null result plainly. The fix this points to — a canonical vocabulary — is tested directly in §5.

What does work as a cross-field join key is the statistics — the same terms recur in every discipline, so they connect papers that share nothing else.

5. Audited statistical terms as cross-field join keys

§4 argued that the missing ingredient is a *canonical vocabulary*, not merely more papers. We therefore tested a frozen 24-term `statistic` vocabulary (`p-value`, `standard-deviation`, `confidence-interval`, `effect-size`, `correlation`, `regression`, `hypothesis-testing`, …). Statistical terms usually have more stable cross-field semantics than topic phrases, but they are not perfectly field-invariant: estimators, tests, parameterisations, and reporting conventions vary. A shared term is therefore a retrieval key, not evidence that two papers used an identical operation or reached a substantively comparable result.

Audit protocol. The unit was a paper–term assignment. The re-audit was **rule-based with LLM-assisted adjudication**: explicit semantic and provenance rules were applied programmatically, and borderline cases were adjudicated against those rules with model assistance. An independent second coder later re-coded a stratified sample of these decisions, yielding $\kappa = 0.767$ (see §7). A term was semantically accepted only when the supplied evidence or locally available article body supported an unambiguous statistical or probabilistic sense. Generated statistics summaries, bibliography-only matches, lexical collisions, non-inferential uses of "significant", class probabilities labelled p, and symbols such as H_0 without an explicit null-hypothesis sense were excluded. The audit resolved all **519** legacy assignments: **420 accepted** and **99 rejected** (19.1%), with no unresolved audit decisions. The tag indicates an accepted occurrence, which may be a focal use or a substantive mention; it does *not* by itself assert that the paper employed that method.

Grounding sensitivity analysis. The legacy corpus did not preserve equally strong provenance for every excerpt. Of the 420 semantically accepted assignments, **272** have an exact normalized excerpt located in the locally stored article body; 97 belong to papers whose source text is not stored locally, 41 excerpts were not located exactly in the extracted body, and 10 supplied excerpts resolve to a reference block even though a separate body occurrence supported the semantic decision. Consequently, the full accepted analysis and the stricter exact-local-body subset are reported side by side. The latter is a provenance check, not an unbiased lower bound, because local source availability is non-random.

Across the full accepted set, at least one statistical key occurs in **100 of 126 papers** and **37 of 39 disciplines**. The strict exact-local-body subset contains **272 assignments in 77 papers across 32 disciplines**. The recalculated term-level results are:

join key	accepted: papers / disciplines	exact local body: papers / disciplines
`mean`	59 / 31	39 / 25
`standard-deviation`	34 / 24	23 / 15
`correlation`	39 / 22	23 / 15
`regression`	33 / 19	24 / 16
`statistical-significance`	27 / 19	18 / 13
`p-value`	24 / 17	10 / 9

The previously reported `p-value` + `standard-deviation` neighbourhood also changes: it contains **10 papers across 10 disciplines** in the accepted analysis, not 14 papers across 13 disciplines. Requiring both excerpts to be exact local-body matches leaves **3 papers across 3 disciplines**. These co-occurrences define candidates for comparison; they do not establish substantive equivalence.

The revised result is narrower than the original claim but still supports the structural point: a controlled namespace produces more cross-field joins than highly specific free-text phenomenon strings. It is evidence for the retrieval design, not proof that statistical vocabulary is field-invariant or that the joins are scientifically novel. The natural next step is a similarly audited canonical `phenomenon` namespace.

A first structural test already supports this. Our priming schema — the association / secondariness / modulation slots — groups **16 papers across 8 disciplines** (neuroscience, cognitive and social psychology, behavioral economics, economics, marketing, education, linguistics) under one canonical phenomenon, even though each field names it differently. Detected on the tag graph, it is an existence proof that a canonical phenomenon layer, grounded in shared *structure* rather than shared *words*, connects fields the free-text facet left disjoint (Fig. 4).

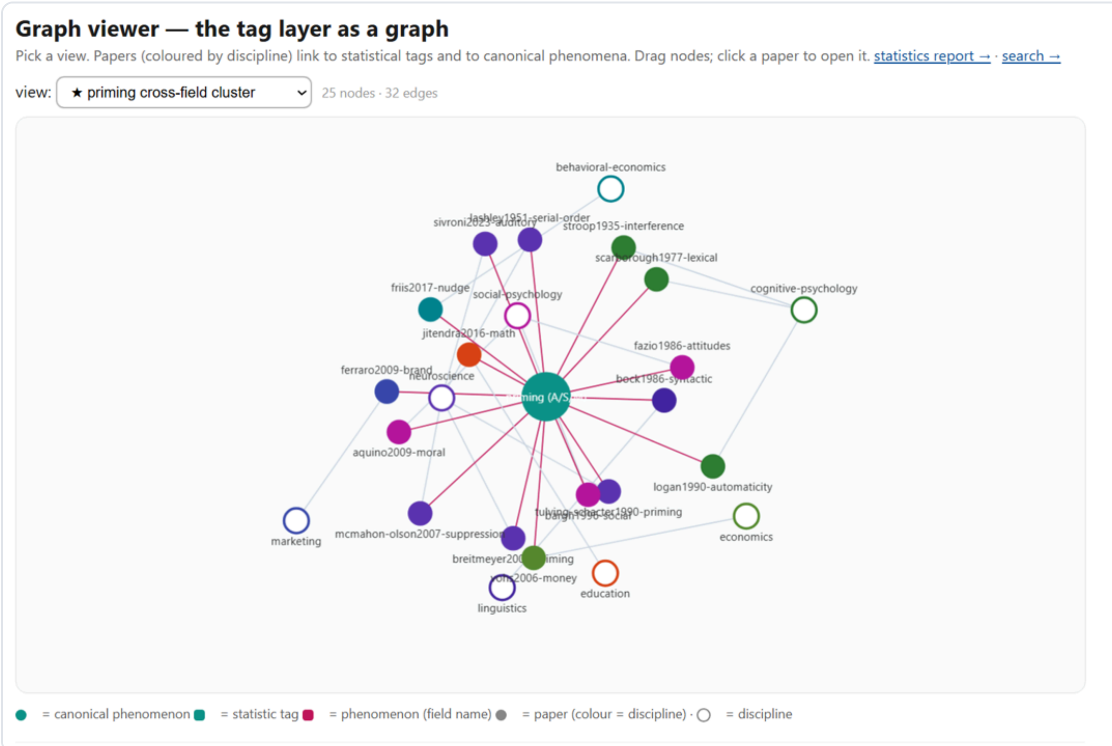


Fig. 4. The priming cross-field cluster in the interactive graph viewer (**real data**): one canonical *priming (A·S·M)* node at the centre links to 16 papers, coloured by discipline, wired out to their eight fields (neuroscience, cognitive and social psychology, economics, behavioral economics, marketing, education, linguistics) — though each field names it differently (interference, automaticity, nudge, money-priming, syntactic priming). 25 nodes · 32 edges. **Click the figure** for the live interactive graph: shir-openu.github.io/meta-tagging-showcase/GRAPH.html (opens locally as `GRAPH.html`).

We are careful about the claim: a shared *method* vocabulary is a weaker signal than a shared substantive finding — it marks where a connection *may* lie, consistent with the compass framing (§6), not a discovery.

Applied to priming across nine disciplines: association and modulation always hold, and only secondariness can fail.

5.1 A grounded test: does each field's priming actually carry the three features?

Showing that "priming recurs across many fields" is trivial and long known; it needs no tool. The non-trivial, *falsifiable* question is whether *each* field's priming phenomenon actually instantiates the three defining features of the meta-priming definition — **A · association** (a prime–target relation), **S · secondariness** (the prime is incidental / prior / task-irrelevant, not the reported object itself), and **M · modulation** (the outcome differs from a baseline) — *grounded verbatim in that paper's own text*. We turned each feature into a checkable, verbatim-anchored tag and audited all 16 priming papers across the 8 disciplines.

Association and modulation are *structural*: they are read directly from the tagged prime→target and baseline→outcome slots, and hold in **16/16** papers. Secondariness is the *discriminating* test — the prime must be textually marked as incidental, prior, or task-irrelevant, and it **can fail**. It is grounded in **14/16** papers; the two exceptions are findings, not noise:

- **Education** (schema-based mathematics instruction): the "prime" is a *deliberately taught* schema — the object of instruction, not an incidental cue — so secondariness is weak by construction.
- **Neuroscience** (serial-order motor control): motor *pre-activation* is a constitutive part of the produced action, not a task-irrelevant prime.

This is the payoff of grounding: the layer does not rubber-stamp the definition, it **locates the boundary** where a field's usage of "priming" stretches it. The verification is a tag intersection — `phenomenon = priming AND feature:association AND feature:secondariness AND feature:modulation` — whose *complement* surfaces the exceptions automatically. In the implemented sorted-postings index, key lookup is expected O(1), intersection costs $\Theta(\Sigma|P_i|)$ plus output, and the universe difference needed for a complement can cost $\Theta(N)$. The query remains practical for thousands of papers when the selected postings are sparse, but it is not independent of corpus size (Fig. 6).

Fig. 5. Grounded A·S·M verification of the 16-paper priming cluster. Association and modulation are read from the structural slots (16/16); secondariness is a textual test that can fail (14/16). **Live interactive version** (per-paper evidence, drill-down): shir-openu.github.io/meta-tagging-showcase/PRIMING_SCALE.html.

What it looks like at scale. The same indexed query can be evaluated over a thousands-paper corpus, with cost determined by the selected postings and output: one canonical *priming* node wired to its three defining features, surrounded by the disciplines that study it, each field coloured by whether all three features hold (turquoise) or one is under-met (red-wine — an exception to investigate). The corpus itself is one graph — papers, tags, disciplines, and canonical phenomena as nodes; grounded tags and cross-field projections as edges — over which such questions become queries rather than manual reading.

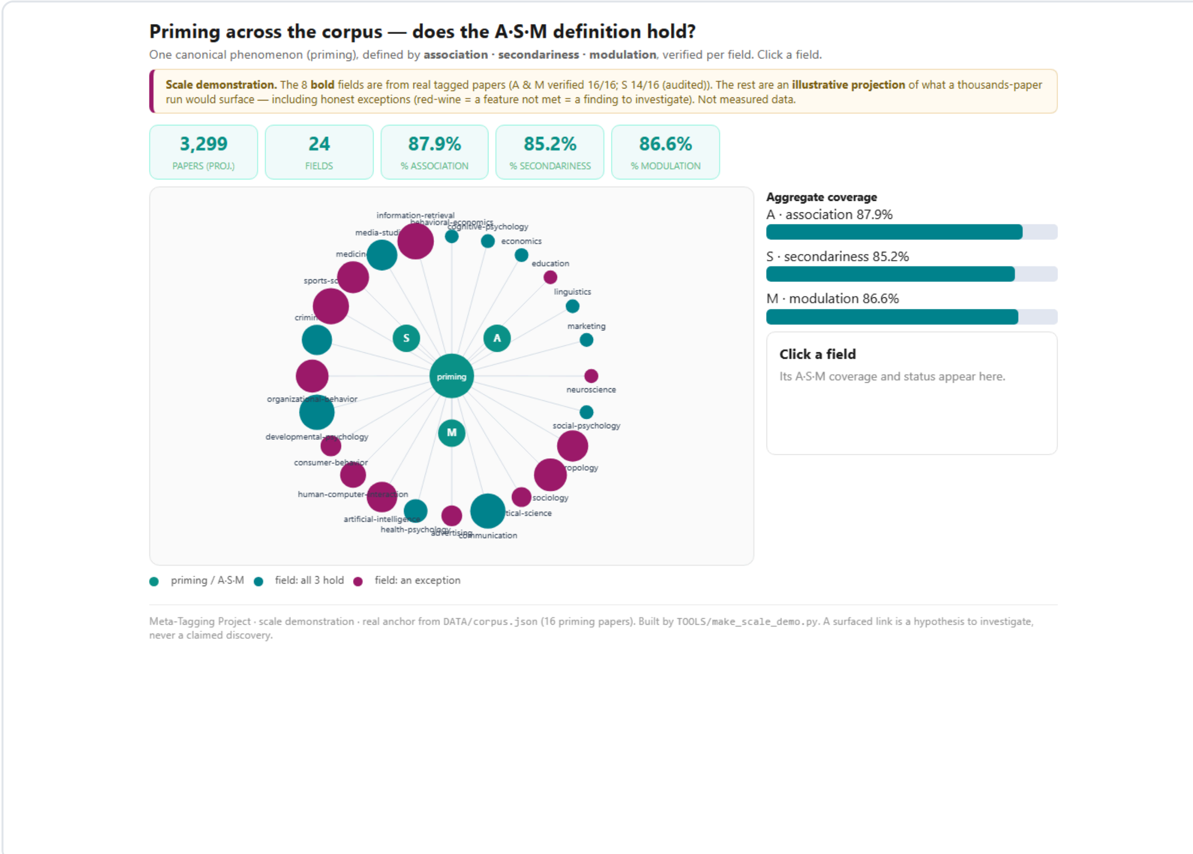


Fig. 6. The same A·S·M verification as an **illustrative projection** to a thousands-paper corpus (24 fields): one canonical *priming* node wired to its three features, each field turquoise if all three hold or red-wine if one is under-met (an exception to investigate). The eight bold fields carry the real audit numbers; the rest illustrate what a large run would surface — *not measured data*. **Click the figure** for the live interactive version: shir-openu.github.io/meta-tagging-showcase/PRIMING_SCALE.html.

A search over the tag layer that runs entirely in the browser, with no server, so the demonstration cannot rot.

5.2 Querying the layer: an implemented static, client-side search

The revised search ships as static assets served over HTTP: a small manifest, a postings dictionary, and lazy-loadable metadata shards (500 records per shard by default). It needs no application server or query-time database, but it is no longer described as one embedded self-contained file. The interface exposes exactly the facets the layer defines. **Input:** the reader composes a query from free text (title / author / key term), the statistics join-keys grouped by family (central tendency, spread, association, inference, advanced), claim-type, and discipline, and chooses `ALL` (intersection) or `ANY` (union) for the selected statistical keys. **Output:** every paper matching the query, *across all fields at once* — each result card shows the paper's discipline, claim-type, the matched tags, its key terms, and a link to the colour-tagged full text with the evidence excerpt (Fig. 7). For one selected facet, dictionary lookup is expected O(1); ALL and ANY queries merge sorted postings in $\Theta(\Sigma|P_i|)$ time plus output work. An unfiltered listing or a common facet can be $\Theta(N)$, and a complement requires a universe difference that can also be $\Theta(N)$. Record shards are fetched only for the current result page. Thus the implemented gain is reduced scanning and transfer for selective queries, not a size-independent asymptotic bound.

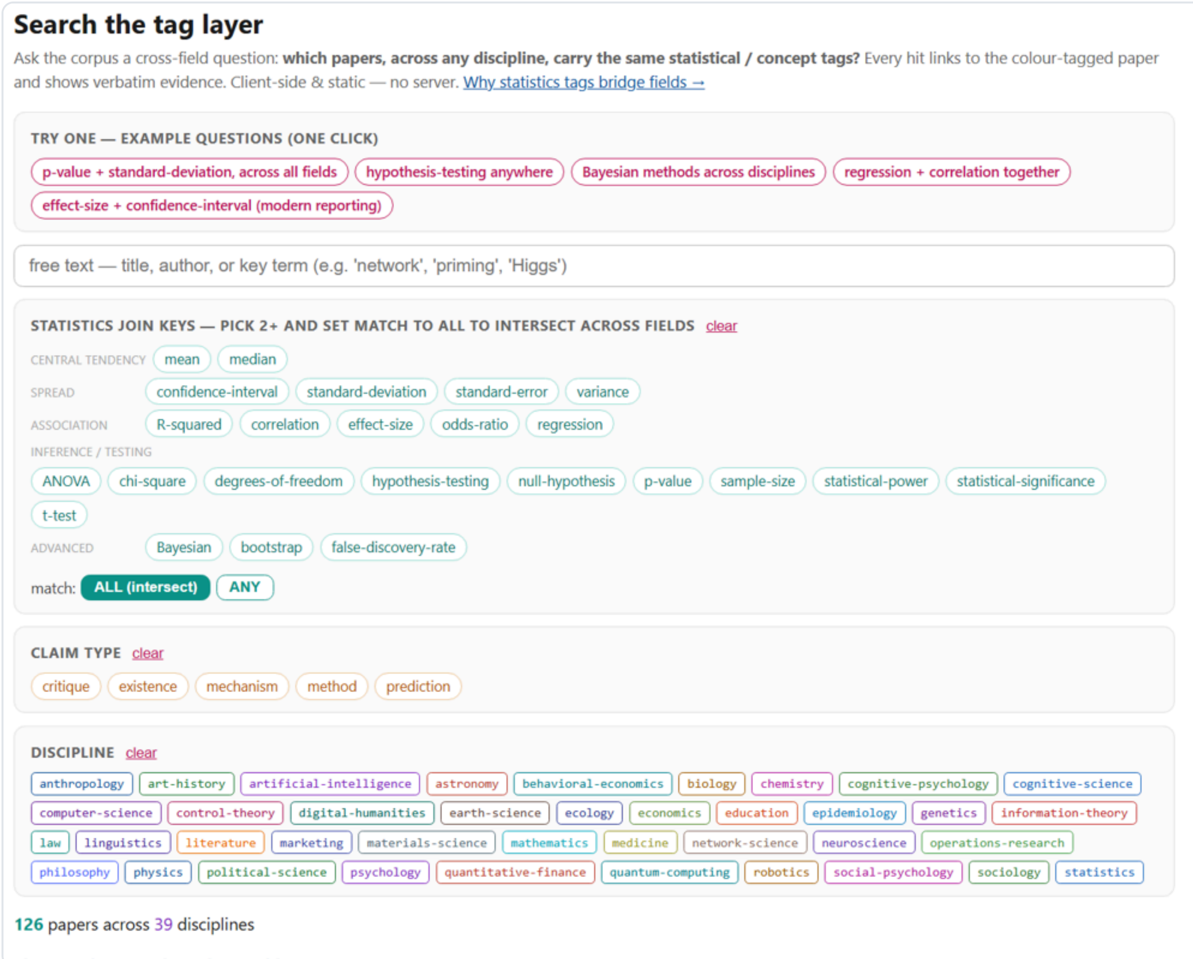


Fig. 7. The static, client-side search. The reader intersects field-neutral tags — statistics join-keys (grouped), claim-type, discipline, free text — with an `ALL` / `ANY` switch; results are papers from any field, each with its matched tags, key terms, and a link to the colour-tagged evidence. No application server; static HTTP hosting is required. **Click the figure** for the live search: shir-openu.github.io/meta-tagging-showcase/SEARCH.html.

The same instrument that fails to find shared names can locate regions the literature occupies but never named — demonstrated for short-term implicit memory.

5.3 The negative space: locating terms the literature lacks

The null result of §4 is usually read as a disappointment: free-text `phenomenon` strings almost never collide across fields, so they fail as join keys. Read the other way round, the same fact is a **detector**. If a set of papers occupies one region of the concept space and yet shares no canonical label, that is not merely a retrieval failure — it is evidence that the literature never coined a term for the region. A layer that can only describe the vocabulary a field already has is a catalogue. A layer that can point at vocabulary a field is *missing* is doing something else, and nothing in a citation graph can do it: co-citation and bibliographic coupling operate on the links between papers, not on the words the papers failed to share.

The procedure. Take two dimensions that the literature already treats as established, cross them, and ask which cells are *populated but unnamed*. Operationally, over the tag layer: find papers that cluster on their tagged attributes yet share no tier-1 key phrase — a populated cluster with no dominant canonical label is a candidate missing term. The test is deliberately conservative: absence of a term is claimed only when the region is demonstrably occupied.

A worked case. The canonical memory taxonomies cross duration against awareness, but not symmetrically. *Implicit* is filed under long-term non-declarative memory — procedural skill and priming — while *short-term* is treated as working memory and tacitly explicit. The cell that would be named **short-term implicit memory** is therefore crowded with data and thin on name — there is no standard box for it in the diagrams, and the phrase itself is barely used. Counting occurrences in titles and abstracts (OpenAlex, July 2026) gives:

Term	Works	Status
short-term memory	155,238	established axis
working memory	112,772	established axis
implicit memory	4,083	established axis
candidate unifying term for the cell		
short-term implicit memory	12	
implicit short-term memory	21	≈52 works in total
implicit working memory	19	
the same cell, under local names		
repetition priming	1,876	
masked priming	1,478	
perceptual priming	457	≈4,350 works in total
echoic memory	367	
short-term priming	172	

Roughly **4,350** works occupy the region under five paradigm- or modality-specific names, while any unifying term for it appears in about **52** — a ratio near **80:1**. The correct claim is therefore *not* that the concept is absent; it plainly is not, since thousands of studies operate there. The claim is that the region is **fragmented**: each paradigm names its own corner, and no superordinate term joins them. That is the same structure this paper documents for priming in §5.1, one level up — and it is why a study of immediate masked repetition and a study of echoic persistence can share a subject matter without sharing a retrieval path. A term such as **short-term implicit memory** would name what they have in common: memory that is measured within seconds and expressed without the subject being asked to remember. Every short-term priming effect in §5.1 depends on such memory, yet no facet of the taxonomies picks it out, which is why a researcher working in one of those paradigms has no name to search for the others by.

What this does and does not establish. It does not show that the missing term *ought* to be adopted; naming is a decision for the field, and a term that unifies at one level may obscure distinctions at another. It shows that the gap is **locatable and countable** rather than a matter of impression, and that the instrument which located it is the same one that failed to find cross-field collisions in §4. One demonstrated case is not a survey: the procedure is offered here as a method with a worked example, and the obvious next step is to run it systematically over the crossed dimensions of a field and report the full set of candidate gaps, including the ones that turn out on inspection to be adequately named already.

Two claims, and they stand or fall separately. What precedes this paragraph is a *worked case*: the region named by **short-term implicit memory** is demonstrably occupied — roughly 4,350 works under five paradigm-specific names against about 52 under any unifying one — and that case rests on its own evidence. It does not depend on how many other unnamed regions exist, and nothing below revises it. What follows is a *separate and weaker* exercise: a sweep asking how often the corpus's own vocabulary is missing from a global authority file. A sweep can come back small without touching a demonstrated case, and here it does.

An external check, and a correction to the size of that second claim. Once the tag vocabulary was reconciled to Wikidata (§8.1), the sweep could be run against a global authority file rather than against a count of our own. The substrate matters here and constrains the result. The `concepts` facet is *analyst-assigned*: a label chosen to describe what a paper is about, so its absence from Wikidata could reflect our wording as much as a gap in the literature, and it cannot carry the argument. The `key_phrases` layer can: it is lifted from the papers themselves, and 988 of 990 checked appear literally in the paper's own text. Of the **176** key phrases used by two or more papers, only **32** are multi-word — single-word canons are mostly lemmatisation artefacts and were excluded. Of those 32, **three** are absent from Wikidata entirely on three independent probes (a 25-result label and alias search, a full-text search, and an English Wikipedia title lookup, since an article implies an item): *lexical decision* (6 papers), *category accessibility* (2) and *repetition suppression* (2). A first pass over the analyst-assigned concepts had suggested fourteen such gaps; seven of those turned out to exist and three more to be project vocabulary rather than the literature's, so the honest figure is the smaller one. The claim in this paragraph is deliberately the reduced version.

A second pattern is sharper than absence. Five further multi-word phrases are reachable only through a redirect to a differently-titled article: *semantic priming*, *zero shot*, *variational inference*, *fault tolerant* and *scale free*. In the priming case, the authority file does not lack the term so much as **refuse the distinction** — *social priming*, *semantic priming* and *conceptual priming* all resolve to the single item *Priming (psychology)*. A layer that records what each paper actually wrote preserves a distinction that reconciliation to a coarser authority would silently collapse, which is an argument for linking out to authorities rather than being replaced by them.

This is a compass, not an answer: it points at where to look, and does not claim to settle what is found there.

6. Framing: a compass, not an answer

The layer is intended to enable *far reading* — a corpus-wide overview in the spirit of distant reading [1] — that indicates *where* a cross-disciplinary connection may lie, so that a scholar knows where to invest the slow *close reading* that can actually establish it. A surfaced link is a hypothesis to investigate, never a claimed discovery; because one cannot prove the absence of a prior connection, every candidate is presented with its literature check, not asserted.

Because reading on demand does not scale, and an ungrounded summary cannot be checked by the person who did not do the reading.

6.1 Why a grounded layer, rather than reading on demand

A language model can read any paper in this corpus and answer questions about it, which invites the objection that a tagging layer is unnecessary machinery. The objection deserves a direct answer, because it determines what such a layer is *for*. It is not for access: these papers were always available. Four properties distinguish a grounded layer from reading on demand.

- **Verifiability, not access.** A model that reads a paper and reports a property of it produces a claim that cannot be checked without redoing the reading. Every assignment here instead carries the sentence that licenses it and an explicit provenance state, so a reader who doubts a judgement can inspect the quoted text and disagree on the record. As machine-generated summaries proliferate, the scarce commodity is not content but provenance for claims about content.
- **Questions about the set, not the paper.** "Which papers, in any field, carry both of these terms?" is not a question one can put to a reader one paper at a time; it is a set operation over an index. Deep reading of a single document and population-level querying are complementary, and only the second needs a layer.
- **Reproducibility.** A layer is computed once and queried repeatedly, returning the same answer each time; re-reading is neither cheap nor deterministic. Every count in this paper regenerates from the stored records, so a disagreement about a number is settled by re-running the build rather than by trusting the author, and a correction propagates to every statement that depends on it.
- **Join keys must be constructed, not assumed.** A reader who knows one field inherits that field's vocabulary, which is precisely the barrier: the terms of §5.3 do not retrieve one another. A model trained on the literature inherits the same vocabulary, and is therefore structurally blind to the words the literature never coined — the case in §5.3 is not one an on-demand reader can be asked to find, because the query term does not exist.

A fifth consideration is human rather than computational. The obstacle facing a researcher is usually not that a relevant literature is inaccessible but that it is *unrecognisable*: a political scientist studying racially coded language has little reason to search a lexical-decision literature. Rendering both in a shared notation makes the common structure visible, and the attached evidence lets a reader judge in seconds whether the connection is worth pursuing. We offer this as the design rationale for the layer, and as a worked demonstration in §5.1, not as a measured claim about human performance; establishing that would require a study of readers, which we have not run.

Stated plainly: a curated corpus, single-coder tagging in places, a measured κ of 0.767 on the statistical layer only, and facets that are not all grounded.

7. Limitations

- The `statistic` facet (§5) records an accepted statistical or probabilistic occurrence, not necessarily a method the paper used. Sharing a key marks a candidate neighbourhood, a much weaker signal than a shared substantive result. The full audit rejected 99 of 519 legacy assignments; semantic judgment error may remain.
- Evidence provenance is incomplete. Only 272 of 420 accepted statistical assignments currently have an exact normalized excerpt located in a locally stored article body. The other provenance states are reported separately and must not be described as exact-body verified.
- Tagging currently depends on a large language model to read prose; the grounding guard bounds but does not eliminate interpretation error.
- The statistical-tag re-audit was rule-based with LLM-assisted adjudication. Its reliability has since been **measured** rather than assumed: an independent second coder (a different AI system, blinded to the original decisions) re-coded a stratified sample of 120 assignments, giving **κ = 0.767** (95% CI 0.652–0.882; 88.3% raw agreement), conventionally read as *substantial*. The disagreement was asymmetric — the second coder accepted 11 assignments this audit rejected and rejected 3 it accepted — so the original coding was the stricter of the two. The 14 disagreements are published for adjudication. A second coder is not a gold standard, and agreement between two systems is weaker evidence than agreement with expert human coders, which remains to be done.
- The corpus is curated for breadth, not sampled representatively; distributions above describe *this* corpus only.
- **Not every facet is verbatim-grounded, and the difference had not been stated before.** The `content_tags` groups carry an `evidence` string and are grounded in it; the `concepts` facet carries a term and a type but no quote, and is an analyst label for what a paper is about. Any argument that rests on the papers' own words must use the first kind, which is why §5.3 was re-run on key phrases and its claim reduced. Giving the `concepts` facet the same grounding requirement is outstanding work.
- The `place` field is semantically underspecified: the TEI export writes it as `<pubPlace>`, but reconciling it showed the values are overwhelmingly the *institutional* city of the authors (DeepMind → London, MIT → Cambridge MA), not the publisher's. Two values (`online`, `Online`) are not places at all. The reconciliation records this rather than silently cleaning it.
- The `study_location` facet (where the participants were, as distinct from where the authors were) is grounded but **thinly populated**: 4 papers of the 38 human-subjects papers whose text is available state it

in a sentence a machine can find. The cap is structural — most human-subjects work in this corpus is the paywalled priming literature, whose text is held internally and never published, so the facet cannot see it. The low rate is reported as a finding about how rarely sample provenance is recoverable from a paper, not as a measurement of the corpus.

- `replication-status` and `limitations` are grounded in each paper's *own* text; external replication outcomes (e.g. well-known failed replications) are deliberately *not* injected, because they are not grounded in the source — a candidate future extension, not a silent edit.
- The phenomenon join key needs canonicalisation before it can surface cross-field links at scale (§4).
- The implemented graph export does not include `ALIAS_OF`, `CONGRUENT_TO`, projected links, or graph-analysis results. The priming projection is a separate curated analysis and Fig. 6 is explicitly illustrative.
- Full-text HTML is unsuitable as the primary Git-backed store for a much larger corpus. Production deployment should keep compact, versioned metadata and build code in version control, put redistributable full text in content-addressed object storage, and serve postings and immutable record shards through static hosting. The current in-memory build is intended for thousands to low tens of thousands of records, not unbounded scale.
- Novelty claims are unfalsifiable in the strict sense; we can search and report, not prove a negative.

TEI, Web Annotation, nanopublications and distant reading each do part of this; what is new is requiring all of it at once.

8. Relation to prior work

The layer sits between text-encoding standards (TEI), stand-off annotation models (the W3C Web Annotation Data Model [2]), machine-readable scientific assertions (nanopublications [3]), scholarly knowledge graphs, and computational distant reading [1]. Its distinguishing commitments are the union of (i) mandatory verbatim grounding of every tag, (ii) facets chosen to be field-neutral join keys rather than topic descriptors, and (iii) a hypothesis-testing stance about whether such a layer can surface undrawn cross-field links — reported here honestly: free-text `phenomenon` strings do *not yet* collide across fields, while an audited controlled vocabulary of statistical terms (§5) creates cross-field retrieval joins. These joins remain semantically and provenance-qualified, pointing the way to canonicalising the rest without claiming field invariance.

A gazetteer reconciles a place name across languages; this layer reconciles a sentence across disciplines — the same operation, so the same standards fit.

8.1 The same operation across languages and across disciplines

The closest working analogue to this layer is not in text mining but in historical gazetteers. MEHDIE, a project linking Hebrew, Arabic and Syriac sources on Middle Eastern history, states the problem it exists to solve in terms that transfer exactly: *"Scholars in the Humanities often strive for depth and rigor in a single field… This specialization and expertise, however, come at the cost of bodies of knowledge becoming isolated from each* *other"* [4]. Its answer is to reconcile a place name written in one *language* to a shared identifier, keeping the source attestation. This layer reconciles a verbatim sentence written in one *discipline* to a shared tag, keeping the quote. **Discipline is to a tag layer what language is to a gazetteer**, and the machinery is the same: a surface form, an authority record, and a pointer back to where the form was found.

Taking that seriously has a concrete consequence, because two existing standards already do the work. Expressed in **SKOS**, each facet becomes a concept scheme and each tag an addressable concept with a URI, labels in more than one language, and an `exactMatch` or `closeMatch` to an external authority. Expressed in the **W3C Web Annotation Data Model** [2], each grounded tag becomes an annotation whose target carries a `TextQuoteSelector` — `exact`, `prefix`, `suffix`. That selector is this project's verbatim-grounding rule, already standardised: what had been a house rule turns out to be a W3C rule, and adopting it costs nothing but makes the layer readable by tools that know nothing about this project.

What the conversion produced, and what it cost. Reconciling the controlled vocabularies against Wikidata succeeded completely for the structural facets and only partly for the substantive ones: **39 of 39** disciplines and **13 of 13** methods were linked, against **15 of 25** audited statistical terms and **61 of 88** concepts shared by two or more papers. A link was asserted only where both a matching label or alias *and* a type statement licensing it could be shown; everything else was recorded as unmatched rather than guessed. The asymmetry is not noise. The further a term sits from the shared scaffolding of research and towards the specific content of a field, the less likely a global authority file is to have it — which is the same gradient this layer was built to work along.

Re-expressing the tags as annotations also re-verified them, independently of this project's own checker. Of **1,964** verbatim spans, **1,313 of the 1,321** whose paper text we are licensed to publish resolve to an exact position in the published page — **99.4%**; the remaining 643 sit on link-only pages where the rights audit removed the body text but kept every tag. Getting there required fixing three things the conversion exposed: some stored quotes join two separate passages with an ellipsis and are not single substrings of anything (the annotation model handles this natively, with one target per span); PDF-extracted text and rendered text disagree about ligatures and whitespace far more often than about words; and our own tag chips, interleaved with the paper's sentences, were cutting quotes in half. None of these were errors in the tags. All three were invisible until a standard was applied to them.

Where the analogy stops, and why the difference matters. MEHDIE reconciles *entities*. Safed is a real place; a proposed match is right or wrong, so the method can be scored — the neighbouring literature reports 0.85 precision and 0.85 recall against a labelled set of 1,500 pairs [5]. This layer reconciles *concepts*, which have no such ground truth: whether two papers mean the same thing by *priming* is exactly what is contested. There is no labelled set to score against, and that is precisely why the layer requires a verbatim quote for every tag and reports an inter-rater κ instead of a precision figure. The two projects are not in conflict; they carry the risk that suits their object. Reconciling places in this corpus made the difference tangible. `Cambridge` matches two of the most important academic cities in the world, and no string test settles which one a paper means; the 42 ambiguous place strings had to be decided by hand with the reason recorded beside each decision, and one string had to be decided per paper rather than once. That is the wall MEHDIE meets and answers with a reconciliation review interface.

A stress test. Nine papers on digital-humanities and Linked-Open-Data infrastructure were tagged with the existing facets — MEHDIE, the Pelagios Network, Recogito, a spatial entity-linkage algorithm, an RDF provenance review, a graph-schema validation paper and three Linked-Open-Data case studies; 45 quotes, each verified against the paper's own text before ingestion. The facets mostly held, with one clean failure: the `method` vocabulary, built for empirical work, had no value for **4 of the 9**. Three of those build systems rather than test hypotheses, and one is a systematic review. The vocabulary is missing *system or tool building*, *systematic review* and *practice report* — a real gap, reported rather than papered over by stretching an existing label. Workshop and tutorial abstracts were excluded from the batch on principle: a syllabus states what will be taught, not what was found, and has no claim to ground a tag in.

A disagreement, kept rather than smoothed over. A second batch of five papers was tagged from the digital-humanities summer school this work comes out of, and one of them contradicts a design decision made here. Münz-Manor and Marienberg-Milikowsky [6] annotated a corpus of late-antique liturgical poetry twice, and report that they began without a tagset at all: *"although there existed a notion of a clear-cut categorization of figurative language in the corpus, no well-defined tagset was created"*, with the devices annotated in *"an entirely flat hierarchy of metaphor, metonymy, synecdoche and simile"*. What visualising the annotations then did was force the categories to change — *"after visualizing the material in this way, it often turns out that the categories themselves need to be re-assessed"* — until categorisation and visualisation formed *"a sort of hermeneutical circle in which both parts influence one another reciprocally"*. Their tagset is an **output** of the analysis.

This layer does the opposite: the facets are fixed in advance and frozen, precisely so that a physics paper and a marketing paper are described by the same slots and can be compared at all. The two positions cannot both be right for the same purpose, and the honest reading is that they answer different questions. A scheme that emerges from a corpus fits that corpus better than any imported scheme could, and is the right instrument for asking what *this* material is doing. A scheme fixed in advance fits no corpus especially well, and is the only kind that can be a join key across corpora that have never been compared. What their result costs us is a real concession, and §4 is already the evidence for it: our frozen `phenomenon` facet did *not* collide across fields, which is exactly the failure mode a category imposed from outside should be expected to produce. What we would say back is that a tagset revised until it fits cannot then be used to measure whether two fields agree — the revision has absorbed the disagreement being measured. Their circle is a virtue for interpretation and a problem for comparison; our freeze is the reverse. Recording that trade-off is more useful than claiming the schema was right.

The same batch supplies an argument this paper should have made earlier. Dekel and Marienberg-Milikowsky [7] attack distant reading at the same joint §1 does, and more precisely: its weakness, they write, *"lies in its exclusive reliance on canonical and authoritative historiographies, one or two for each national literature, something which is bound to over-simplify the complexities of national literatures"*. Their answer is to replace the authoritative summary with the aggregated readings of many people. Ours is to replace it with what each paper itself says, quoted. Both are refusals of the same shortcut.

The convergence worth recording is empirical, not rhetorical. MEHDIE reports that general gazetteers *"contain few, if any, toponyms in the Semitic scripts relevant to the sources of Middle Eastern History"*, that *"Even Wikidata, which is essentially multilingual, shows low recall"*, and gives a specific case: Wikidata holds a modern Hebrew variant of a toponym but not the archaic spelling the historical source actually uses [4]. Section 5.3 reports the same shape of gap on a different axis. Their conclusion — a preference for a bottom-up, intra-domain strategy over a centralised one — is an argument for building domain-grounded layers that *link out* to central authorities rather than waiting for those authorities to cover the domain. That is what this layer now does.

Everything is public and rebuildable: corpus, code, vocabulary, annotations, and a rights manifest saying what may be redistributed and what may not.

9. Availability

A public showcase provides the report and, for the papers under a verified redistributable licence, the full colour-tagged text; every other paper shows the tag layer with short licensing excerpts and a link to the source, not the reproduced text. Redistribution rights are verified per paper (against the source landing page and the Crossref API) in an explicit `rights_manifest.json`; being free-to-read is not treated as permission to redistribute. The repository adds a canonical corpus JSON, an explicit audit-decision file, invariant validation of every record, sorted postings, lazy metadata shards, a typed adjacency export, and one build command. **Reproducibility** here means identical *computations*, not byte-identical files: the build regenerates the inverted index, graph export and validation deterministically from the resolved corpus and passes a 7-test regression suite. The statistical-grounding audit (`DATA/section5_statistics.json`) is a **frozen, human-adjudicated artifact** computed against each paper's full text; because most full text is not redistributed (rights), those grounding sub-counts are shipped as a fixed input rather than recomputed from the redistributed pages. It includes a **static, client-side search** over the tag layer and a generated statistics audit report. Live showcase: shir-openu.github.io/meta-tagging-showcase/.

Linked Open Data. The layer is also published in two standards it did not invent, at [/lod/](/lod/): a SKOS vocabulary (`vocab.jsonld`, `vocab.ttl`, and a browsable `vocab.html` carrying the same graph as embedded JSON-LD), the tags as a W3C Web Annotation collection (`annotations.jsonld`), and the corpus as records (`corpus.jsonld`) — together about 47,800 triples, each file parsed and counted as part of the build. Two things are deliberately absent. The reliability layer is private and is never exported. And paper-level facets are `dcterms:subject` statements rather than annotations, because they have no verbatim quote behind them and only tags that do become annotations; blurring that line would overstate what is grounded. The reconciliation carries its own evidence: every asserted link records the label or alias that matched and the type statement that licensed it, and the place decisions that required human adjudication are kept, with their reasons, in `DATA/place_adjudication.json`. A companion paper applies the same layer to a single phenomenon, testing a meta-disciplinary definition of priming against 38 tagged papers in 9 disciplines; the shared notation it uses is also implemented as the data model of an open experiment builder (PrimingToolbox, source), in which a design-time check warns when a configured experiment violates the definition — so the same schema that describes published papers also constrains experiments not yet run. The inter-rater materials (coding manual, blinded sheet) are published at `kappa-study/`. Archived on Zenodo — DOI: 10.5281/zenodo.21432188.

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7. O. Münz-Manor, I. Marienberg-Milikowsky, "Visualization of Categorization: How to See the Wood and the Trees," *Digital Humanities Quarterly* 17(3), 2023 — a tagset that emerged from the analysis rather than preceding it.
8. Y. Dekel, I. Marienberg-Milikowsky, "From Distant to Public Reading. The (Hebrew) Novel in the Eyes of Many," *magazén* 2(2), 225–252, 2021, doi:10.30687/mag/2724-3923/2021/01/003 — distant reading without the authoritative history.

Preprint · v11 linked-open-data revision · July 2026 · licensed CC BY 4.0 · DOI: 10.5281/zenodo.21432188 · cite via CITATION.cff in the project repository. The v8 audit re-audited all 519 legacy statistical assignments, recalculated §5, separated semantic acceptance from exact-local-body grounding, implemented a sharded sorted-postings search and reproducible build pipeline, exported only supported graph relations, and corrected the architecture and complexity claims. v9 adds a per-paper redistribution-rights manifest, states the audit method (rule-based with LLM-assisted adjudication) and the absence of an inter-rater study, and reconciles all release metadata. Earlier versions introduced the priming A·S·M analysis and the statistical join-key experiment. v11 reconciles the controlled vocabularies to Wikidata, publishes the layer as SKOS and W3C Web Annotation (§8.1), re-verifies 99.4% of verbatim spans by anchoring them as TextQuoteSelectors, adds nine digital-humanities papers as a facet stress test, adds a grounded study-location facet, and reduces the §5.3 negative-space claim to what the papers' own words support. v17 tags five papers from the digital-humanities summer school this work comes out of, and records a methodological disagreement with one of them rather than smoothing it over: their tagset emerged from the analysis, ours is frozen in advance, and §8.1 states what each choice buys and costs. v19 replaces every paragraph with one sentence written to say what that paragraph argues, and puts the paragraph itself one click behind it; the figures and their legends are always shown, tables open on request, and the summaries are kept in a separate file so that a paragraph whose text changes loses its summary rather than keeping a stale one.